COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212	**SECTION:**	CPE31S2
DATE PERFORMED:	09/02/26	**DATE SUBMITTED:**	09/02/26
NAME:	Ivan Wayne San Juan	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 5.1 Implement Ansible roles in playbooks

1. PROCEDURE OUTPUT

Task 1: Use when command for different distributions

1.1

```
kaizohost@kaizohost-VirtualBox:~$ ls
ansible-lab      Desktop    Downloads  my-repo   Public  Templates
CPE212_SanJuan   Documents  Music      Pictures  snap    Videos
kaizohost@kaizohost-VirtualBox:~$ cd CPE212_SanJuan
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan$ git pull
Already up to date.
```

When I issued git pull, Git connected to the remote repository, authenticated me, and checked for new changes but found none, so it printed "Already up to date." No actual file transfers or merges occurred because my local repository was already fully synchronized with the remote branch.

1.2

inventory.yaml

```yaml
all:
  children:
    ubuntu:
      hosts:
        server1:
          ansible_host: 192.168.56.102
          ansible_user: kaizoserver1
          os_type: Ubuntu


        server2:
          ansible_host: 192.168.56.103
          ansible_user: kaizoserver2
          os_type: Ubuntu

    centos:
      hosts:
        server3:
          ansible_host: 192.168.56.105
          ansible_user: server
          os_type: CentOS

    linuxservers:
        children:
          ubuntu:
          centos:
```

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook -K -b install_apache.yml
BECOME password:

PLAY [all] ***********************************************************************************************************************************************

TASK [Gathering Facts] ***********************************************************************************************************************************
ok: [server2]
ok: [server3]
ok: [server1]

TASK [update repository index] ***************************************************************************************************************************
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
fatal: [server3]: FAILED! => {"changed": false, "cmd": "apt-get update", "msg": "[Errno 2] No such file or directory: b'apt-get'", "rc": 2, "stderr"
"", "stderr_lines": [], "stdout": "", "stdout_lines": []}
changed: [server2]
changed: [server1]

TASK [install apache2 package] ***************************************************************************************************************************
ok: [server1]
ok: [server2]

TASK [add PHP support for apache] ************************************************************************************************************************
ok: [server2]
ok: [server1]

PLAY RECAP ***********************************************************************************************************************************************
server1                    : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server3                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0

kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$
```

1.3

install_apache.yml

```yaml
---
- hosts: all
  become: yes
  tasks:

    - name: update repository index
      apt:
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: install apache2 package
      apt:
        name: apache2
        state: present
      when: ansible_distribution == "Ubuntu"

    - name: add PHP support for apache
      apt:
        name: libapache2-mod-php
      when: ansible_distribution == "Ubuntu"
```

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook -K -b install_apache.yml
BECOME password:

PLAY [all] ***********************************************************************************************************************************************************

TASK [Gathering Facts] ***********************************************************************************************************************************************
ok: [server2]
ok: [server1]
ok: [server3]

TASK [update repository index] ***************************************************************************************************************************************
skipping: [server3]
changed: [server2]
changed: [server1]

TASK [install apache2 package] ***************************************************************************************************************************************
skipping: [server3]
ok: [server1]
ok: [server2]

TASK [add PHP support for apache] ************************************************************************************************************************************
skipping: [server3]
ok: [server1]
ok: [server2]

PLAY RECAP ***********************************************************************************************************************************************************
server1                    : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server3                    : ok=1    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
```

Server 3 was skipped because the script uses apt, whereas CentOS requires dnf.

1.4

```yaml
- name: install apache2 package
  apt:
    name: apache2
    state: present
  when: ansible_distribution == "Ubuntu"

- name: add PHP support for apache
  apt:
    name: libapache2-mod-php
  when: ansible_distribution == "Ubuntu"

- name: update repository index
  dnf:
    update_cache: yes
  when: ansible_distribution == "CentOS"

- name: install apache2 package
  dnf:
    name: httpd
    state: latest
  when: ansible_distribution == "CentOS"

- name: add PHP support for apache
  dnf:
    name: php
    state: latest
  when: ansible_distribution == "CentOS"
```

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook -K -b install_apache.yml
BECOME password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [server1]
ok: [server3]
ok: [server2]

TASK [update repository index] *************************************************************************************
skipping: [server3]
changed: [server1]
changed: [server2]

TASK [install apache2 package] *************************************************************************************
skipping: [server3]
ok: [server1]
ok: [server2]

TASK [add PHP support for apache] **********************************************************************************
skipping: [server3]
ok: [server1]
ok: [server2]

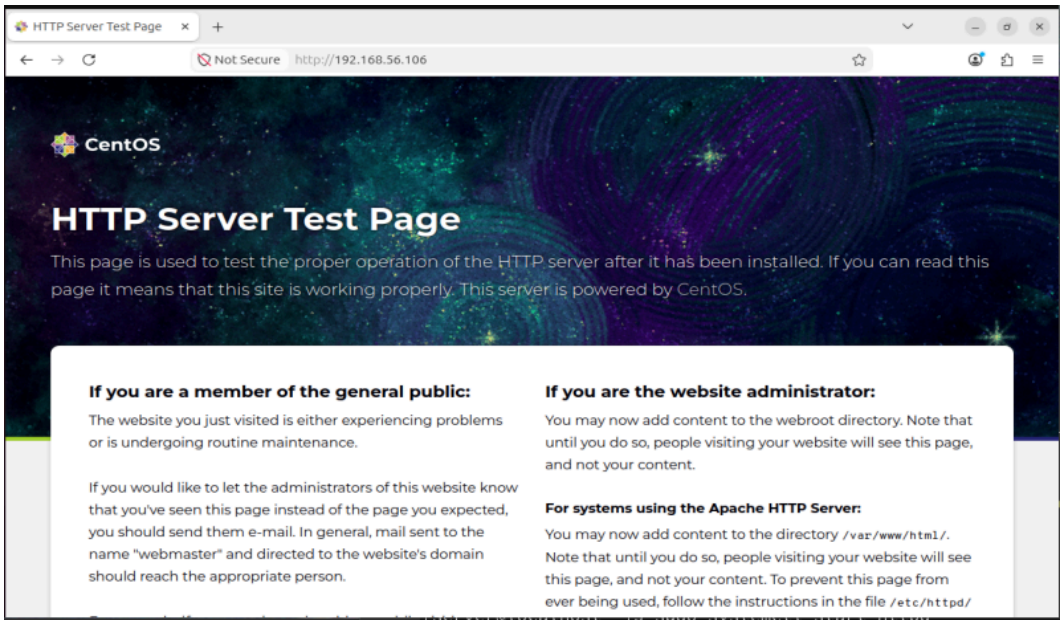
TASK [update repository index] *************************************************************************************
skipping: [server1]
skipping: [server2]
ok: [server3]

TASK [install apache2 package] *************************************************************************************
skipping: [server1]
skipping: [server2]
changed: [server3]

TASK [add PHP support for apache] **********************************************************************************
skipping: [server1]
skipping: [server2]
changed: [server3]

PLAY RECAP *********************************************************************************************************
server1                    : ok=4    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
server2                    : ok=4    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
server3                    : ok=4    changed=2    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
```

5.1



2.1

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook -K -b install_apache.yml
BECOME password:

PLAY [all] *********************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
ok: [server1]
ok: [server2]
ok: [server3]

TASK [update repository index] *************************************************************************************************************************
skipping: [server3]
changed: [server2]
changed: [server1]

TASK [install apache2 package and php packages for Ubuntu] *********************************************************************************************
skipping: [server3]
ok: [server2]
ok: [server1]

TASK [update repository index] *************************************************************************************************************************
skipping: [server1]
skipping: [server2]
ok: [server3]

TASK [install apache2 and php packages for CentOS] *****************************************************************************************************
skipping: [server1]
skipping: [server2]
ok: [server3]

PLAY RECAP *********************************************************************************************************************************************
server1                    : ok=3    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
server2                    : ok=3    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
server3                    : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

install_apache.yml

```yaml
---
- hosts: all
  become: yes
  tasks:

    - name: update repository index
      apt:
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: install apache2 package and php packag
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
      when: ansible_distribution == "Ubuntu"

    - name: update repository index
      dnf:
        update_cache: yes
      when: ansible_distribution == "CentOS"

    - name: install apache2 and php packages for C
      dnf:
        name:
          - httpd
          - php
        state: latest
      when: ansible_distribution == "CentOS"
```

2.2



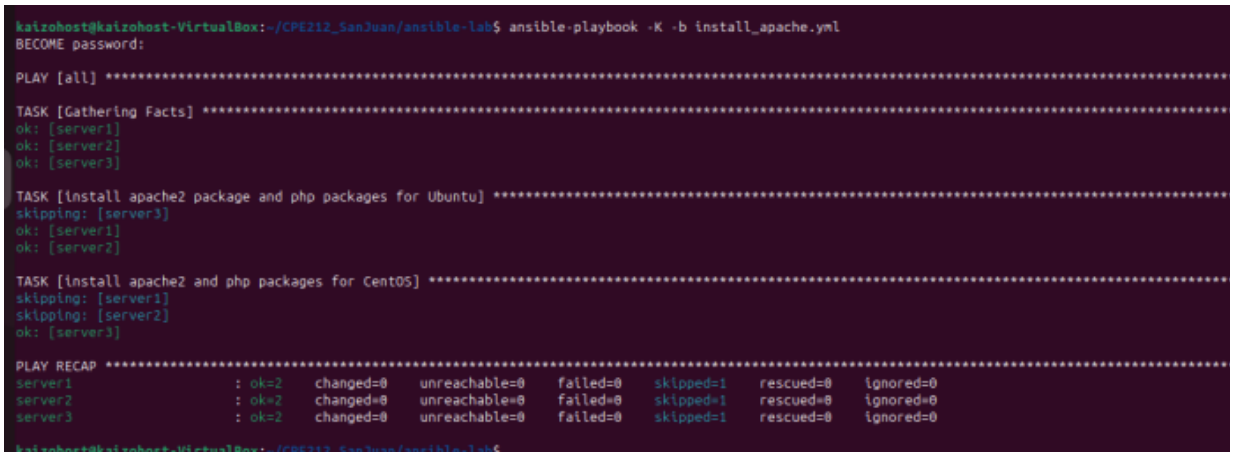
```
install_apache.yml
---
- hosts: all
  become: yes
  tasks:

#    - name: update repository index
#      apt:
#        update_cache: yes
#      when: ansible_distribution == "Ubuntu"

    - name: install apache2 package and php packag
      apt:
        name:
        - apache2
        - libapache2-mod-php
        state: latest
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

#    - name: update repository index
#      dnf:
#        update_cache: yes
#      when: ansible_distribution == "CentOS"

    - name: install apache2 and php packages for C
      dnf:
        name:
        - httpd
        - php
        state: latest
        update_cache: yesS
      when: ansible_distribution == "CentOS"
```



```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook -K -b install_apache.yml
BECOME password:

PLAY [all] ***********************************************************************************************************************************************

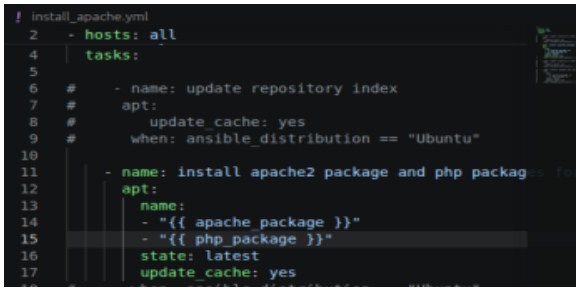
TASK [Gathering Facts] ***********************************************************************************************************************************
ok: [server1]
ok: [server2]
ok: [server3]

TASK [install apache2 package and php packages for Ubuntu] ***********************************************************************************************
skipping: [server3]
ok: [server1]
ok: [server2]

TASK [install apache2 and php packages for CentOS] *******************************************************************************************************
skipping: [server1]
skipping: [server2]
ok: [server3]

PLAY RECAP ***********************************************************************************************************************************************
server1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
server2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
server3                    : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

2.3



```
install_apache.yml
- hosts: all
  tasks:

#    - name: update repository index
#      apt:
#        update_cache: yes
#      when: ansible_distribution == "Ubuntu"

    - name: install apache2 package and php packag
      apt:
        name:
        - "{{ apache_package }}"
        - "{{ php_package }}"
        state: latest
        update_cache: yes
```

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook -K -b install_apache.yml
BECOME password:

PLAY [all] *********************************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************************
ok: [server1]
ok: [server2]
ok: [server3]

TASK [install apache2 package and php packages for Ubuntu] *********************************************************************************************************
fatal: [server1]: FAILED! => {"msg": "The task includes an option with an undefined variable. The error was: 'apache_package' is undefined. 'apache_package' is undefined\n\nThe error appears to be in '/home/kaizohost/CPE212_SanJuan/ansible-lab/install_apache.yml': line 11, column 7, but may\nbe elsewhere in the file depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n\n    - name: install apache2 package and php packages for Ubuntu\n      ^ here\n"}
fatal: [server2]: FAILED! => {"msg": "The task includes an option with an undefined variable. The error was: 'apache_package' is undefined. 'apache_package' is undefined\n\nThe error appears to be in '/home/kaizohost/CPE212_SanJuan/ansible-lab/install_apache.yml': line 11, column 7, but may\nbe elsewhere in the file depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n\n    - name: install apache2 package and php packages for Ubuntu\n      ^ here\n"}
fatal: [server3]: FAILED! => {"msg": "The task includes an option with an undefined variable. The error was: 'apache_package' is undefined. 'apache_package' is undefined\n\nThe error appears to be in '/home/kaizohost/CPE212_SanJuan/ansible-lab/install_apache.yml': line 11, column 7, but may\nbe elsewhere in the file depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n\n    - name: install apache2 package and php packages for Ubuntu\n      ^ here\n"}

PLAY RECAP *********************************************************************************************************************************************************
server1                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
server2                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
server3                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
```

2.4

ansible.cfg | install_apache.yml | inventory.yaml | playbook

inventory.yaml

```yaml
all:
  children:
    ubuntu:
      hosts:
        server1:
          ansible_host: 192.168.56.102
          ansible_user: kaizoserver1
          os_type: Ubuntu
          apache_package = apache2
          php_package = libapache2-mod-php


        server2:
          ansible_host: 192.168.56.103
          ansible_user: kaizoserver2
          os_type: Ubuntu
          apache_package = apache2
          php_package = libapache2-mod-php

    centos:
      hosts:
        server3:
          ansible_host: 192.168.56.106
          ansible_user: server
          os_type: CentOS
          apache_package = httpd
          php_package = php

    linuxservers:
        children:
          ubuntu:
          centos:
```

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ ansible-playbook -K -b install_apache.yml
BECOME password:

PLAY [all] ********************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************
ok: [server2]
ok: [server1]
ok: [server3]

TASK [install apache2 package and php packages for Ubuntu] ********************************************************
ok: [server1]
ok: [server2]
ok: [server3]

PLAY RECAP ********************************************************************************************************
server1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server3                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

CONCLUSION:

This activity showed how Ansible playbooks can be refactored to efficiently manage multiple operating systems, using conditionals, variables, and the `package` module to handle Ubuntu and CentOS within a single, streamlined playbook. It reinforced how reducing redundancy in automation scripts improves maintainability while still achieving the same result across different environments.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.



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